

Analyzing Socio-Political Reasoning Using Multi-Agent AI Trained on Societal Dilemmas

Jeffrey Chen, Micah Chiang, Nikhil Muthukumar, Rohan Adusumilli, Harish Senthilkumar, Chiyan Loo, Sofia Shah, Phil Mui
Department of Computer Science & Engineering, Aspiring Scholars Directed Research Program, Fremont, California

Abstract

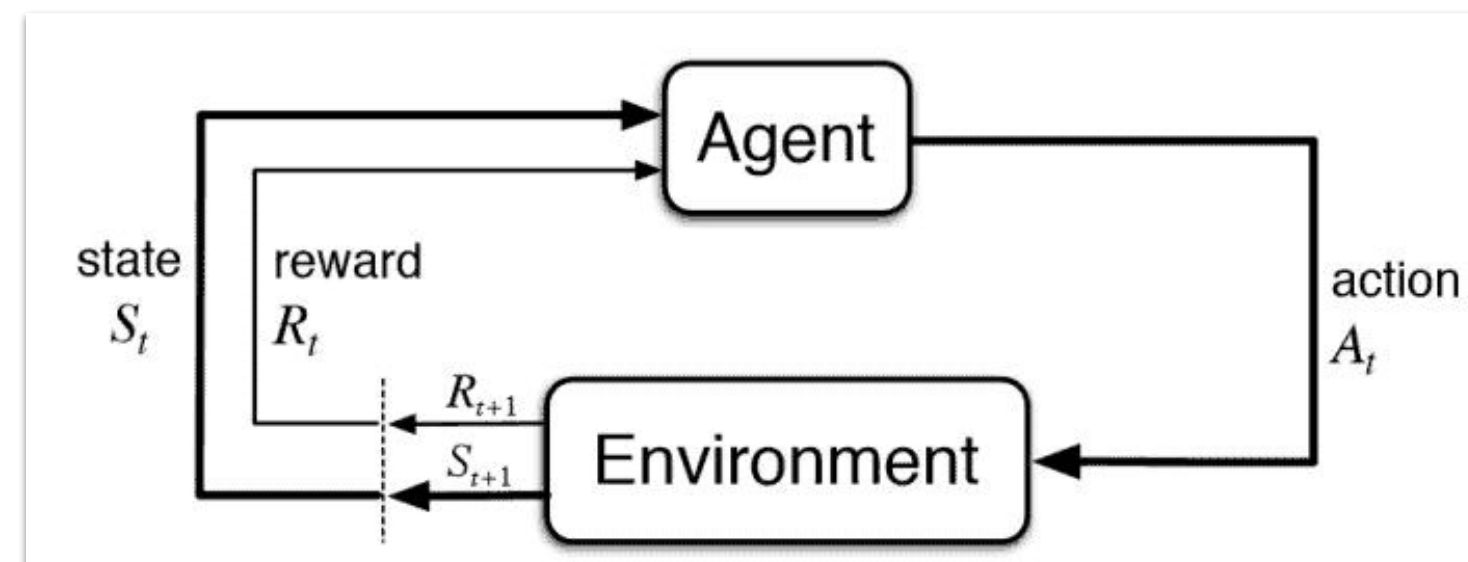
The development and application of Large Language Models (LLMs) have demonstrated exceptional capabilities in generating and responding to text, yet they continue to face limitations in understanding human behavior, particularly in autonomous reasoning and developing insightful analysis about a variety of complex social situations. LLMs lack the ability to independently approach such situations with critical thinking and limited human intervention, along with being less dynamic and autonomous. This problem is important, as it significantly limits the scope and advancement of artificial intelligence based reasoning. In this work, we offer an agent-based simulation framework that models different Stochastic games such as the Prisoner's Dilemma and Diner's Dilemma, where agents have distinct behavioral traits and make decisions under dynamic game settings. Each dilemma is implemented into multiple rounds of simulations, where the collective benefit of each individual action is calculated with the Q-value. Along with the usage of the Q-learning reward function, the agents make decisions based on the highest long-term or short term cumulative reward of each decision. This process is then repeated continuously to mimic reinforcement learning, helping the agents to learn and adapt to the different scenarios and theories which govern the frameworks of the different dilemmas. Overall, our findings reaffirm established simulation patterns such as defection and equilibrium states, showing that the optimization of self-interest often leads to flaws and imbalance due to conflict. Ultimately, our multi-agent framework presents a more accurate representation of societal interactions, offering an intuitive commentary on human behavior and its implications.

Introduction

This project builds a modular multi-agent reinforcement learning framework that turns familiar games into sequential interactions where strategies can learn and adapt. We study four settings in parallel. In the **Repeated Prisoner's Dilemma**, agents face the classic tension between cooperation and defection over many rounds. In **Stackelberg game**, a leader moves first and a follower responds after observing that move, highlighting commitment and anticipation. In the **Diner's Dilemma**, individuals choose private add-ons while the group shares the bill, creating a cost-sharing externality. In a **Senate-style public-goods and coordination game**, agents vote under majority or supermajority thresholds and may bargain to assemble a winning coalition.

Learning is handled with tabular **Q-learning** and a short-horizon n-step variant (Q3). Each agent maintains values for state-action pairs and updates them from experience using temporal difference targets, while exploration rules such as epsilon-greedy or softmax balance trying new actions with exploiting known good ones. **Coalitions** are modeled generically using **Multi Agent Reinforcement Learning (MARL)**: agents can propose agreements that specify target behaviors, simple transfer terms, and responses to violations, and they decide to join or exit based on expected value. This common toolkit lets us compare how learning and coordination mechanisms operate across the four environments without changing the core code, and to study when cooperation, commitment, or bargaining is most effective.

Methodology



Reinforcement Learning: Agents learn a policy by interacting with an environment to maximize long-run reward.

Loop: Observe state $S_t \rightarrow$ take action $A_t \rightarrow$ get reward R_t , next state $S_{t+1} \rightarrow$ update.

Q-learning is the simplest, model-free recipe for turning those experiences into estimates of which actions are best. An agent keeps a table $Q(s,a)$ that estimates how good action a is in state s . After acting and observing the outcome, it updates with

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

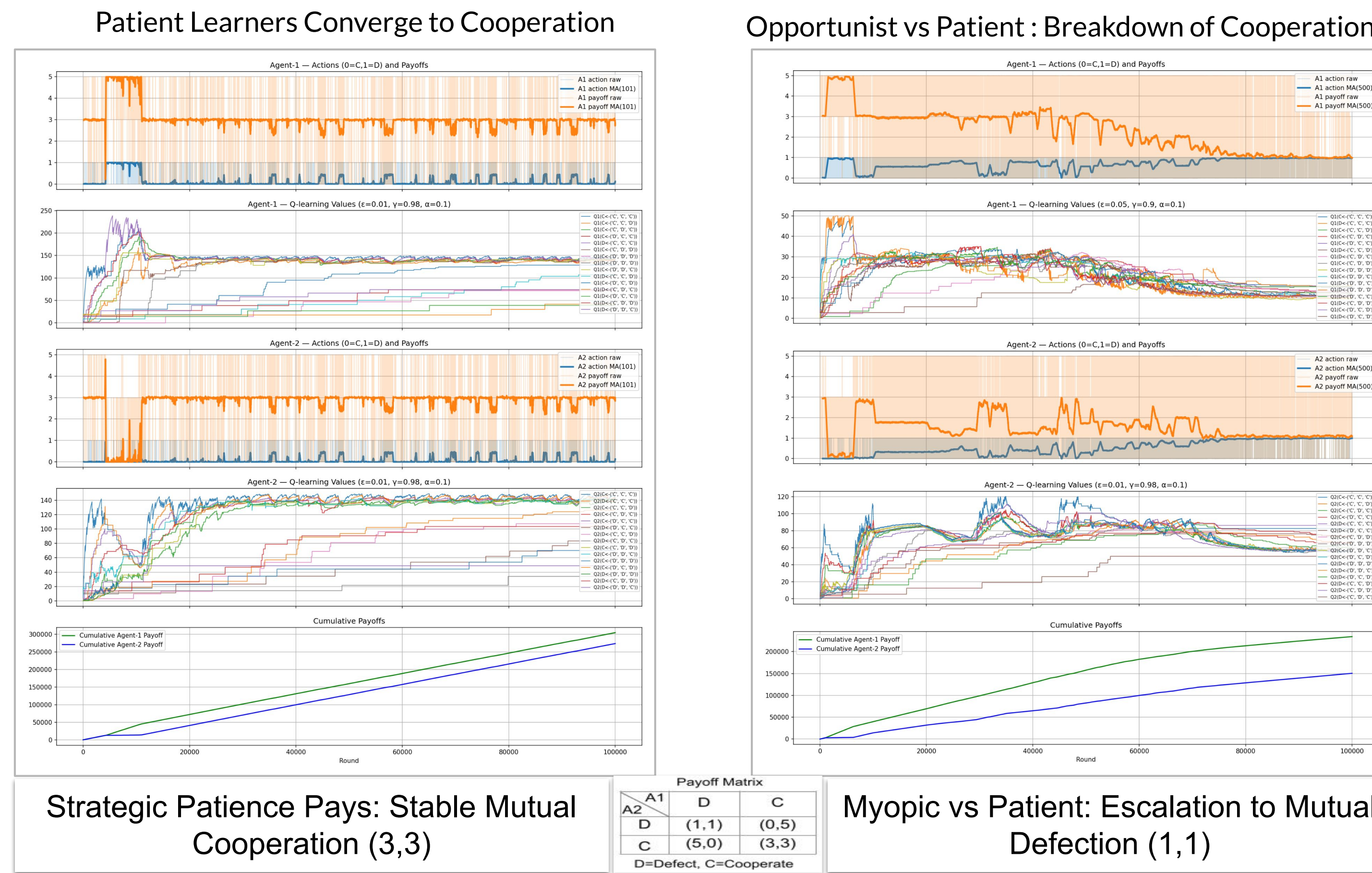
Here r is the **immediate reward** from the environment, s' is the **next state**, and $\max Q(s',a')$ is the agent's **best forecast of future return** if it plays optimally from s' . The bracketed term is the **prediction error** between the old estimate and new evidence;

- α : **learning rate**, sets how policy shifts in response to outcomes
- γ : **discount factor**, controls how much the future matters vs short term gains

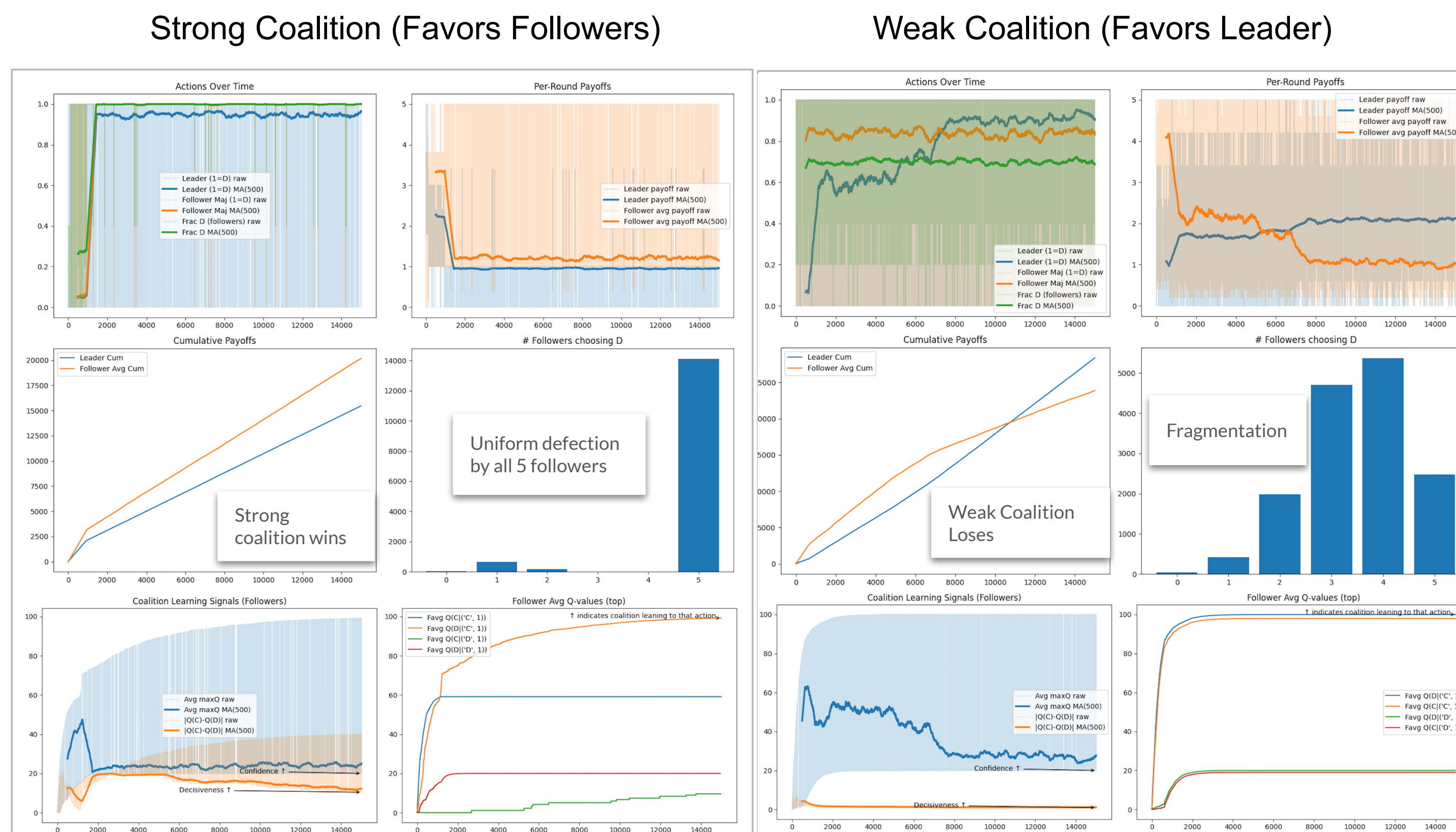
Actions are usually chosen with **ϵ -greedy** exploration—try a random action with probability ϵ

Simulation Results

Repeated Prisoner's Dilemma (Q3 Agents)



Stackelberg Game with MARL (Coalition)



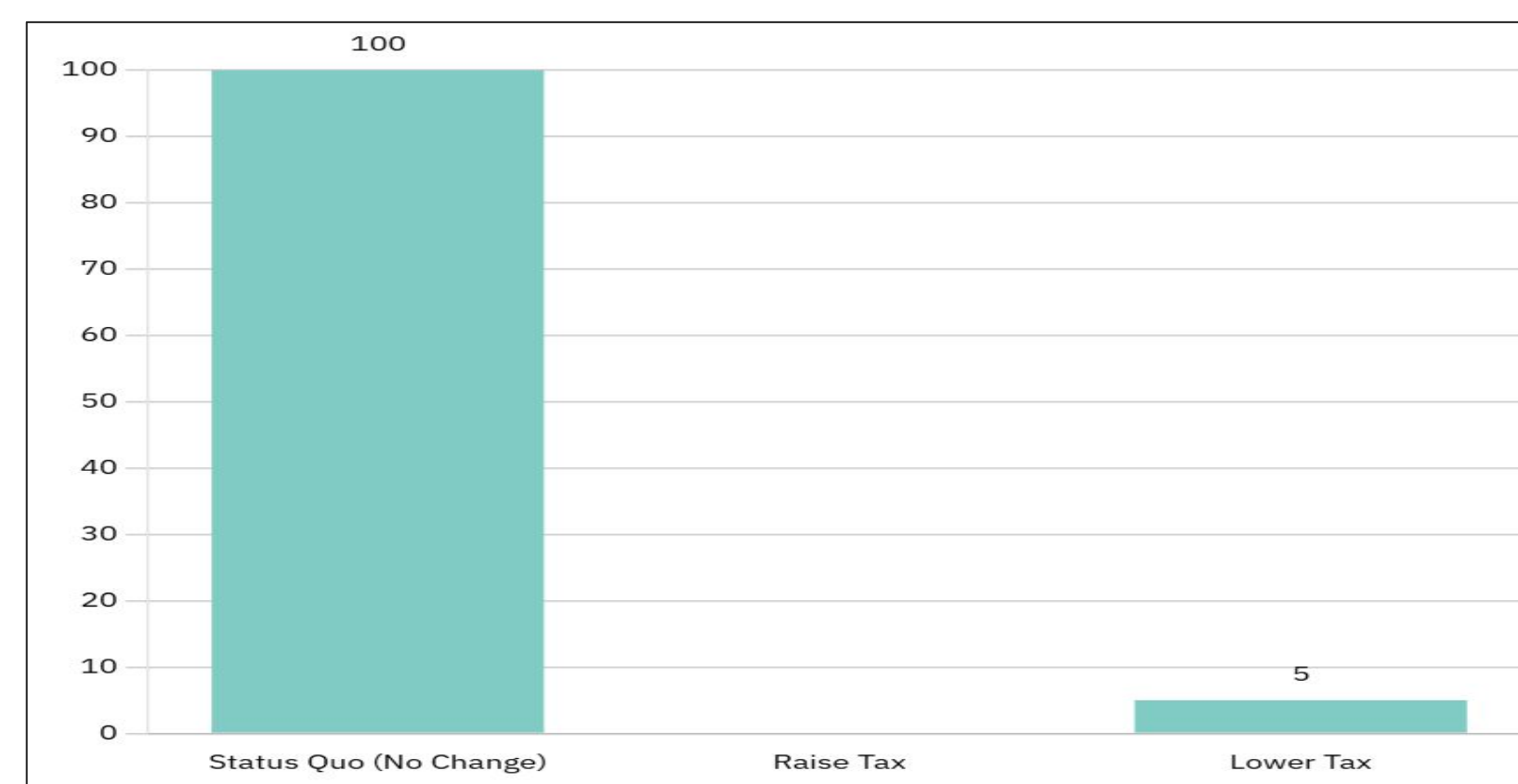
Stackelberg : Leader moves first, and each follower then picks an action with **ϵ -greedy Q-learning**.

- After followers propose actions, a **coalition step** gives them a chance to opt in to coordinate
- Agents join when they're **uncertain** (Q-values are close) or when they're **confident** (low- ϵ)
- Among those who opt in, we compute a majority action; any opt-in agent adopts that majority. Non-participants keep their original choice.

Senate Based Game : Voting Policy

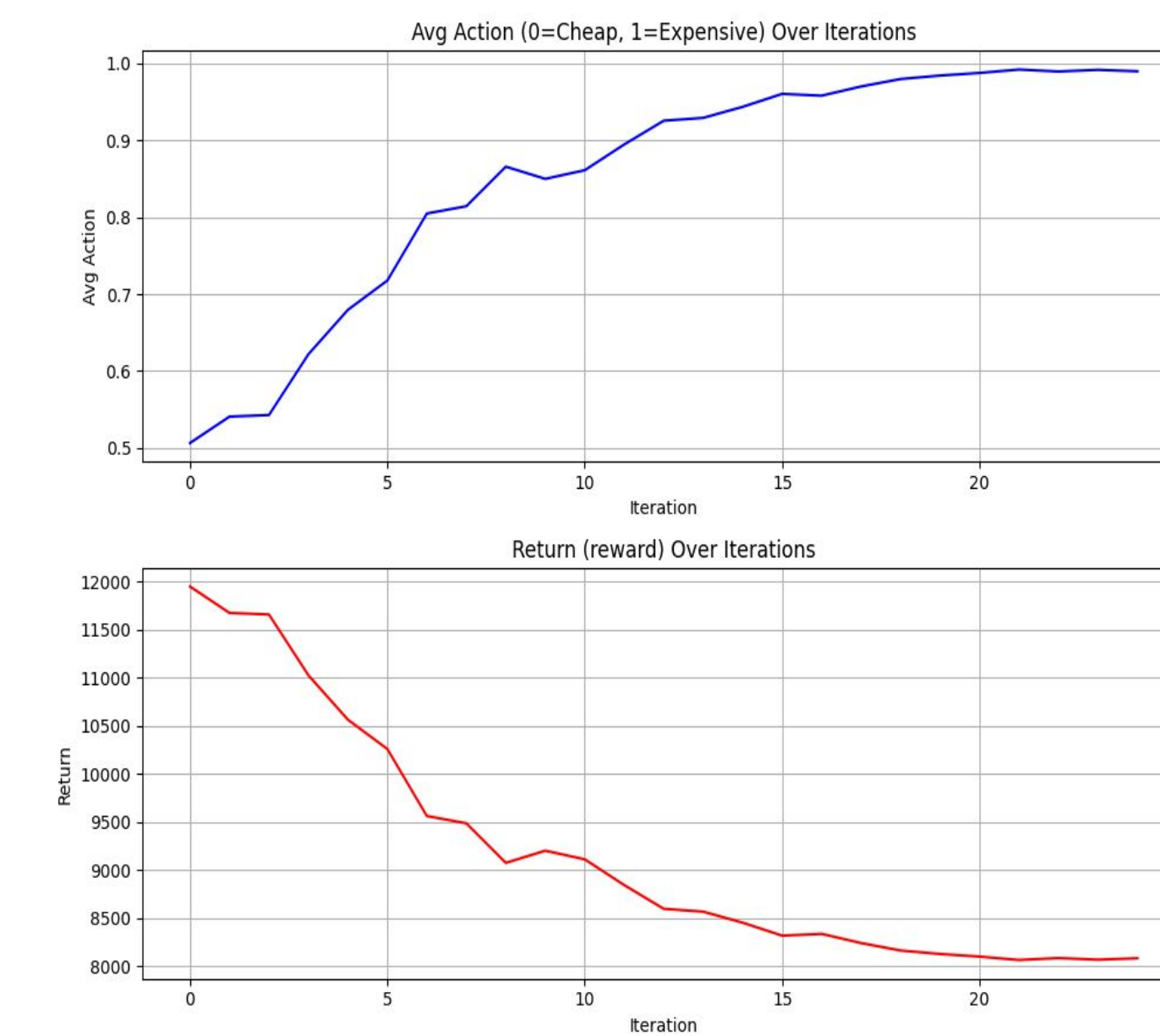
Senate-based game:

- Senators choose whether to support a bill that funds a public good under majority/supermajority thresholds,
- Creating a **coordination/assurance** problem (support is best if enough others also support).



Simulation Results

Diner's Dilemma



Diner's Dilemma:

- When a group splits the bill evenly, each person has an incentive to order something pricier, because most of the extra cost is shared by others.
- Individually rational choices over time (upgrading your order) add up to a **worse outcome for everyone** (a bigger total bill with no matching gain), making it a classic social-dilemma with cost-sharing externalities.

Conclusions

Our results show how adding reinforcement learning to static games changes strategic outcomes. In the Repeated Prisoner's Dilemma, Q3 learning shifts play from the static Nash D,D (1,1) to sustained C,C (3,3) through brief punishment and quick forgiveness. In Stackelberg timing, adding learning and an opt-in coalition layer can erode the leader's advantage; coordinated followers often match or beat the leader once they settle on a common response. The Diner's setting highlights a limitation: with equal bill splitting, agents still over-order and we see no clear gain from learning, pointing to the need for stronger incentives or alternative state features. However, it showed that the agents achieved stable Nash equilibrium through ordering expensive dishes, highlighting the mutually assured destruction outcome and aggressive agentic tendencies. The Senate voting model remains basic but promising as a testbed for coalitions and threshold effects. Overall, the framework turns static models into dynamic studies of socio-political behavior, allowing multi-agent AI to compare policy regimes on equal footing and generate actionable insights.

Future Work

We will link multiple dilemmas into one setting so agents must balance competing incentives across contexts. We will add randomness such as policy shifts, demand spikes, and resource drops to test how robust strategies are under uncertainty. We will also add more agentic traits and actions within each dilemma, encouraging adaptation and modeling real-life learning. To ground the results, we will calibrate to real-world economic and ecological data and compare simulated outcomes to observed patterns. On the learning side, we will extend beyond tabular updates with n-step and Q3 variants, introduce RNN and DRQN for memory, refine state features, and check convergence against Nash and subgame-perfect equilibria.

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